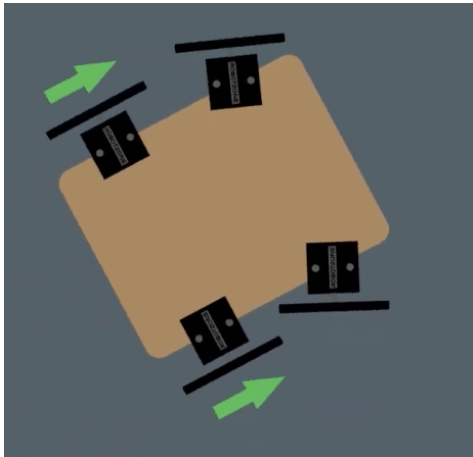
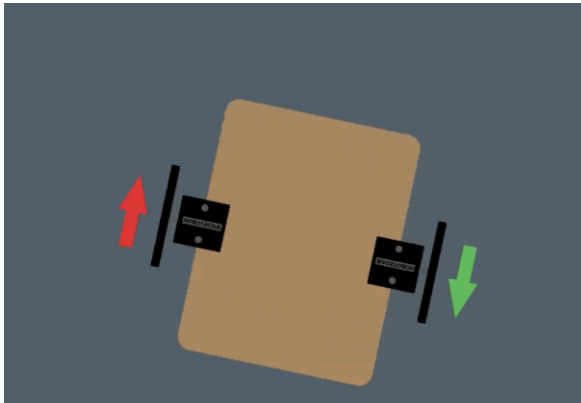


Steering methods

Ackerman steering - conventional steering with

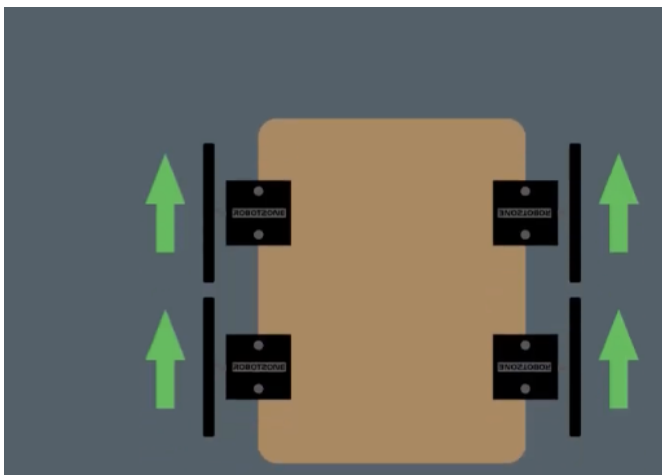


Differential steering



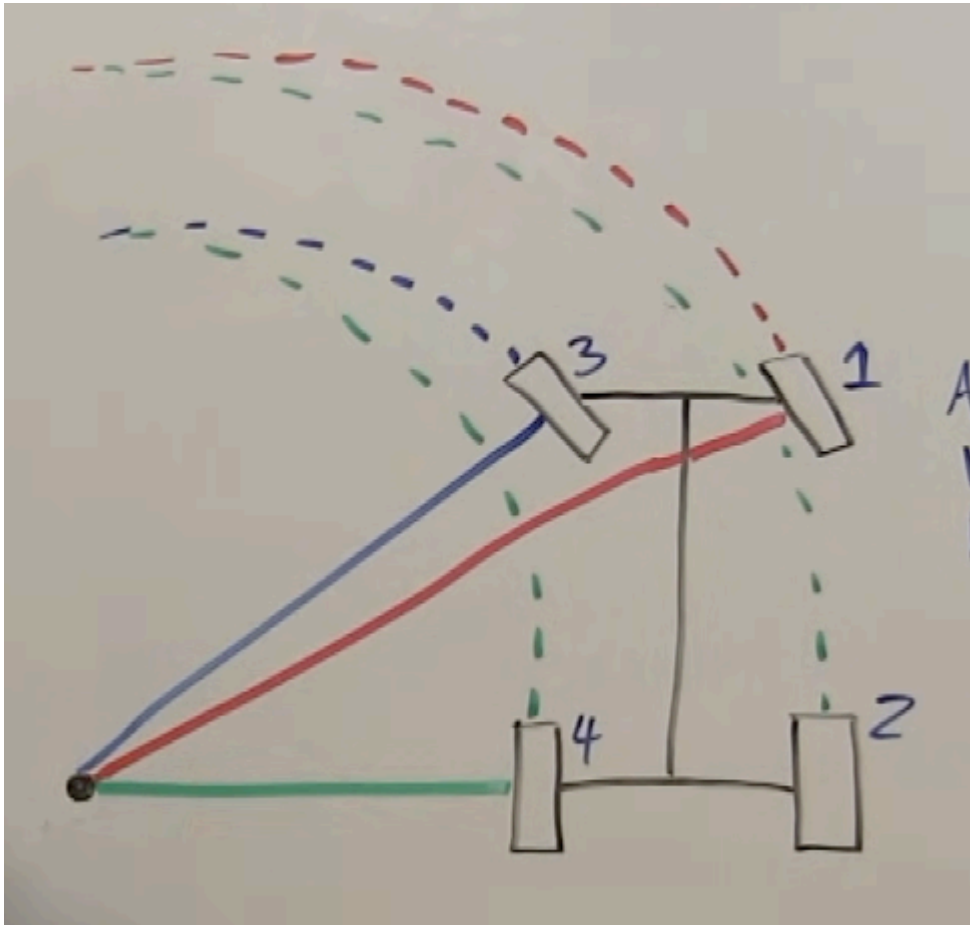
It is used to change the direction of the robot with having differential mechanism

Skid steering - - - the rate at which the left side spins compared to the right side will determine the vector of ur robot — tank



One side wheels is in the same direction, where we can change other side wheels as a whole, all the front and rear other side wheels

Ackerman steering - - - it prevents tyres from slipping - - - which means that the left tyre should not slip



Angle with left side tyre(3) is greater than the outer right tyre (1)

Angular velocity is $1 > 2 > 3 > 4$...

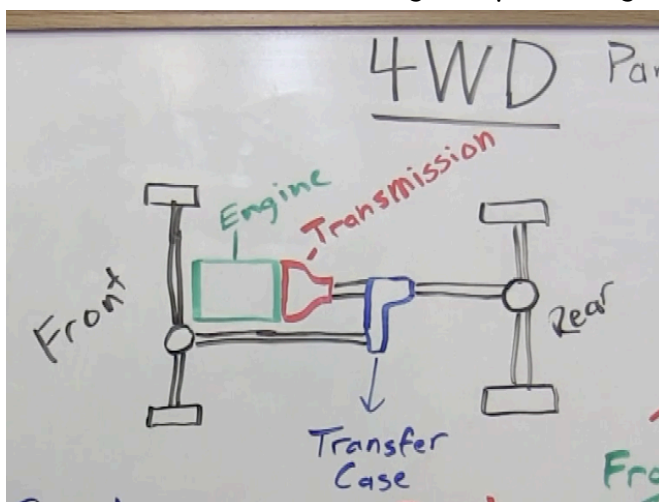
Front wheel, like 1 is moving faster than rear tyre

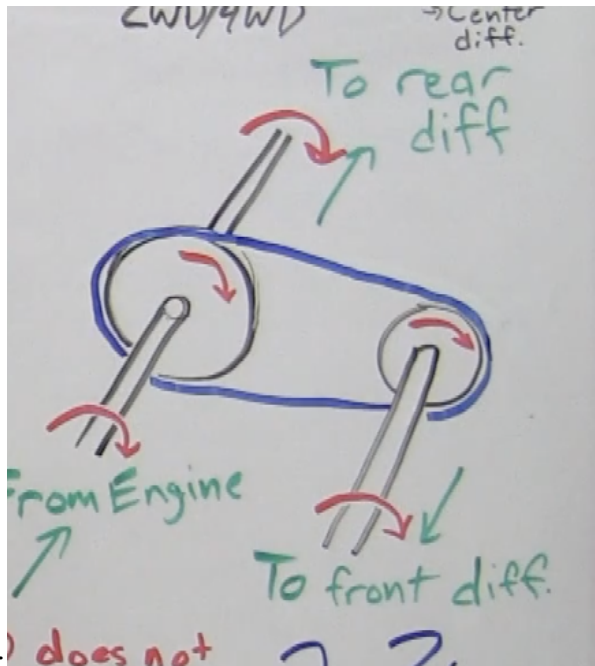
When u turned that's why we need a differential to separate the speed from front tyre and rear tyre

Otherwise, it costs buckling

It rotates at different rates

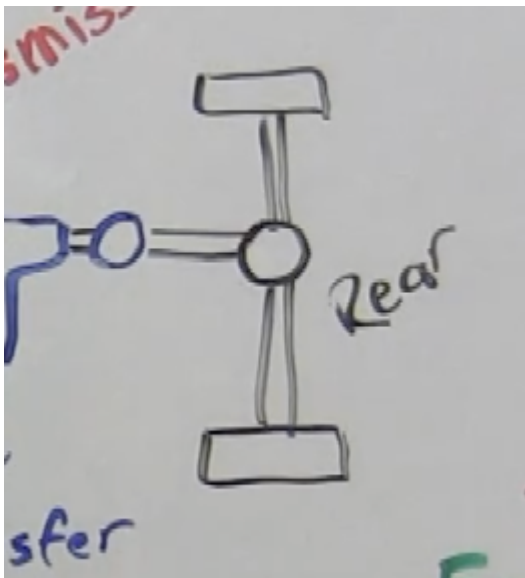
NOW HOW 4 W D WORKS: good point is great off-road traction





It splits the power between these two front wheels
It allows them to rotate at different speeds

Why differential in between rear wheels side



It allows this rotating shaft to rotate at different speed that other rotating shaft point with different rotate at different speed

Not to have under steer

Bad side is if ur using part time 4w wheel and u go on a **paved road** thats have a transmission winded up where the front tyre want to move faster than they can go around a turn , get understeer at tyre wear .

How this transfer case switch between low and high gear
 And how this transfer case switch between just the rear driving and the front driving ???

